

Rui(Ray) Qiu

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RESEARCH FOCUS

Data Visualization, Human-centered Visual Design, Deep learning, LLM-based agents, Multi-modal LLM.

EXPERIENCE

- **Brookhaven National Lab** August 2024 - September 2024
Visiting Scholar
 - Research on applied explainable AI (XAI) techniques to analyze data attribution and interpret the model's reasoning, with a focus on explaining implicit neural representations derived from training data, enhancing transparency in decision-making.
- **The Ohio State University** Jan 2021 - Now
Graduate Research Associate
 - Researched on developing visualization and visual mapping algorithms to visualize corpus space with multiple objectives; Designed effective visual layouts and intuitive interactions to support domain-specific information-seeking needs.
 - Designed and built a visual programming-based system to modularize the literature review process, incorporating intuitive visualization and interaction for a streamlined user experience.
 - Researched on multi-modal language model integrating text, image, and gene modalities to support biology domain scientists in answering complex research questions.
 - Research on explainable AI, analyzing data attribution and investigating factors influencing model-generated answers, improving transparency and interpretability of large language models.
- **The Ohio State University** August 2021 - May 2024
Instructor - Graduate Teaching Assistant
 - Instructor of CSE 2221 Software Design I (core class of CSE undergraduate) for seven semesters. Responsible for teaching lectures, hosting labs and office hours. Received average SEI 4.6/5.
- **Institute for Informatics, Washington University School of Medicine** August 2019 - August 2020
Research Intern & Research Engineer
 - Led development and research on the VoiceConnect project, building a multi-hop interaction language model and enhancing system security in a DynamoDB environment for clinician supply retrieval.
 - Designed and implemented memory mechanisms for a voice assistant's question-answer system using session variables in the VoiceConnect-S Project.
 - Developed the [DoxCompass \(https://doxcompass.web.app/\)](https://doxcompass.web.app/) knowledge retrieval system for systematic review, integrating keyword extraction, clustering, and data visualization, while optimizing performance on Google Cloud with a modularized backend.

EDUCATION

- **The Ohio State University** August 2020 - Now
Ph.D. in Computed Science (Visualization and Computer Graphics)
Columbus, Ohio
 - GPA: 3.90/4.00
- **Washington University in St. Louis** July 2018 - Dec 2019
Master of Science in Finance: Quantitative Finance
St. Louis, Missouri
 - Grade: 3.86/4.00
- **Tianjin University of Finance and Economics** August 2014 - June 2018
Bachelor of Science in Mathematics
Tianjin, China
 - GPA: 3.85/4.00

PUBLICATIONS

[1]] **Qiu, Rui**, Yamei Tu, Po-Yin Yen, and Han-Wei Shen. "VADIS: A Visual Analytics Pipeline for Dynamic Document Representation and Information-Seeking." IEEE Transactions on Visualization and Computer Graphics (2024). (**Best Paper Award at IEEE VIS 2024**)

[2]] Tu, Yamei, **Rui Qiu**, and Han-Wei Shen. "KG-PRE-view: Democratizing a TVCG Knowledge Graph through Visual Explorations." In 2024 IEEE 17th Pacific Visualization Conference (PacificVis), pp. 162-171. IEEE, 2024.

[3]] Tu, Yamei, Xiaoqi Wang, **Rui Qiu**, Han-Wei Shen, Michelle Miller, Jinmeng Rao, Song Gao et al. "An interactive knowledge and learning environment in smart foodsheds." IEEE Computer Graphics and Applications 43, no. 3 (2023): 36-47.

[4]] **Qiu, Rui**, Yamei Tu, Yu-Shuen Wang, Po-Yin Yen, and Han-Wei Shen. "DocFlow: A visual analytics system for question-based document retrieval and categorization." IEEE Transactions on Visualization and Computer Graphics 30, no. 2 (2022): 1533-1548.

[5]] Tu, Yamei, **Rui Qiu**, Yu-Shuen Wang, Po-Yin Yen, and Han-Wei Shen. "PhraseMap: Attention-based keyphrases recommendation for information seeking." IEEE Transactions on Visualization and Computer Graphics 30, no. 3 (2022): 1787-1802.

[6]] **Rui Qiu**. Research on the Warning Model of Health Care Fraud Based on Frequent Pattern Mining Algorithm", Industrial & Science Tribune 16.17, 2017:62-64.

SKILLS

- **Programming Languages:** Python, Java, JavaScript, C++
- **Visualization and Front-end Technologies:** D3, Vue, WebGL
- **Machine Learning& Deep Learning:** Pytorch

HONORS AND AWARDS

- **Best Paper Award at IEEE VIS 2024** Oct 2024
The IEEE Visualization Conference 2024 (IEEE VIS 2024)
- **Graduate Teaching Award** May 2023
The Ohio State University
 - Brief description of the award and its significance